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Paper 1 of 1 — PROCESS DYNAMICS AND CONTROL COURSE CODE: CHE 512

Level 500 • First Semester • Handout

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF CHEMICAL ENGINEERING

LECTURE NOTES

COURSE TITLE: PROCESS DYNAMICS AND CONTROL

COURSE CODE: CHE 512

By

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CHE 512 – Process Dynamics and Process Control

COURSE CONTENT

1. Introduction to process dynamics and control;
2. Review of mathematical tools needed for modelling and simulation,
3. Process dynamics, review of Laplace transforms;
4. Transient behaviour of 1st, 2nd and higher order systems;
5. Process control, Basic components of the process control system and their characteristics, closed loop transient response,
6. Stability analysis/Frequency response Analysis;
7. Design of feedback controllers based on frequency response criteria;
8. Design of model based controllers such as IMC, DMC etc.

1.0 Introduction to Process Dynamics and Control

1.1 Dynamics of Processes

Process dynamics is the time dependent behaviour of a process that predicts the response of the system modelled through differential equations. The objectives of process dynamics are:

- To gain appreciation for the time dependent responses of thermal engineering systems
- To learn construct realistic but reasonable mathematical models of the system
- To study the techniques of solving the resulting differential equations
- To become familiar with control techniques, stability theory, controller design and system design

The behaviour with no controllers in the system is called the open-loop response. The dynamic behaviour with feed-back controllers included with the process is called the closed-loop response. The best way to

illustrate what process dynamics is, is to take some examples

Example 1

Consider a tank containing a perfectly mixed liquid, into which flows a liquid stream as shown in Figure 1.1. The volumetric flow rate into the tank is F_o (m³/s) with a density of ρ_o (kg/m³). The volumetric hold-up of liquid in the tank is V (m³) and the density of the mixed liquid in the tank is ρ (kg/m³). The volumetric flowrate of liquid out of the tank is F (m³/s). The density of the outlet liquid is the same as the density of the tank content. Write a continuity equation describing the dynamics of the system.

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$F_o(t)$

$\rho_o(t)$

ρ

V

(t)

F

(t)

Figure 1.1 Perfectly mixed tank

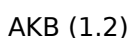
Since the above system is of a definite size it is a macroscopic system
The mass balance is around the whole tank and not just a small differential element inside the tank

$$F_o \rho_o - F \rho = \frac{\partial (\rho V)}{\partial t} \quad (1.1)$$

i.e. $\frac{\partial (\rho V)}{\partial t}$

Example 2

Assuming the tank of example is a CSTR containing component A that reacts irreversibly to form component B per the following reaction



Let the concentration of component A in the inflowing feed stream be C_{Ao} and that in the reactor be C_A , assuming a simple first order reaction. The rate of consumption of reactant A per unit volume will be directly proportional to the instantaneous consumption of A in the tank.

$F_o(t)$

C_{Ao}

(t)

$C_A(t) V(t)$

F

C

(t)

A

(t)

Figure 1.2 CSTR with first order reaction

Let

Flow of A into the system =

$F_0 C_{A0}$

Flow of A out of the system = FC_A

Rate of Formation of A = $-V k C_A$

Total mass balance:

In-out –depletion ($V k$ reaction) = Accumulation

(1.3)

Component A:

3

$F_0 C_{A0} - F C_A - V k C_A = \frac{\partial}{\partial t} (V C_A)$

$\frac{\partial}{\partial t}$ (1.4)

Component B:

$F_0 C_{B0} - F C_B + V k C_A = \frac{\partial}{\partial t} (V C_B)$

$\frac{\partial}{\partial t}$ (1.5)

But

$C_A + C_B = 1$ (1.6)

For this system we have two components and therefore two independent material balances i.e.

Total material balance + 1 component balance, or two component balances. Normally the first option is used.

Example 3: Series of isothermal constant hold-up CSTR

In the system sketched below, product B is produced and reactant A is consumed in each of the three perfectly mixed CSTR by first order reactions occurring in the liquid. Assume that the time constants τ 's and volumes of the three tanks can be different, but the τ 's and liquid volumes are assumed constant so is ρ . The mixture is binary.

F_0

C

(t) A_0

(t) $K_1 K_2 K_3$

$C_{A1} C_{A1} C_{A2} V_2 C_{A2} C_{A2} V_3 F$

V_1

(t) C

(t)

(t) (t) (t) (t) (t) (t) (t) A_3

(t)

Figure 1.3 Constant isothermal holdups CSTR in series

Assume that volume and density of each tank is constant then,

Total balance:

Tank 1:

$$\frac{\partial \rho V}{\partial t} = \rho F_1 - \rho F_2 = F_1 - F_2$$

(1.6)

TANK 2:

$$\frac{\partial V_2}{\partial t}$$

$$= F_1 - F_2$$

(1.7)

TANK 3:

$$\frac{\partial V_3}{\partial t}$$

$$= F_2 - F_3$$

$$(1.8)$$

Component Balance

TANK 1:

$$\frac{\partial C_A}{\partial t}$$

$$V_1 = F (C_{Ao} - C_{A1}) - V_1 k_1 C_{A1}$$

(1.9)

TANK 2:

$$\frac{\partial C_A}{\partial t}$$

$$V_2 = F (C_{A1} - C_{A2}) - V_2 k_2 C_{A2}$$

(1.10)

TANK 3:

$$\frac{\partial C_A}{\partial t}$$

$$V_3 = F (C_{A2} - C_{A3}) - V_3 k_3 C_{A3}$$

(1.11)

These three first order non linear ODE are the mathematical model of the system. The parameters that must be known are V_1 , V_2 , V_3 , k_1 , k_2 , k_3 . The variable that must be specified before these equations can be solved are F and C_A . They can vary with time, but they must be known (given as functions of time). Hence they are called FORCING FUNCTIONS. In addition, the initial conditions of the three concentrations must be known.

Assuming F is constant and the hold-ups and τ 's are the same in all three tanks, then the equation becomes

∂C_A

$\partial t ()$

$+ k + 1\tau C_A^3 = 1\tau C_A^0$

(1.12)

Where:

V

$\tau =$

F

Assignment

5

A fluid of constant density (ρ) is poured into a cone shaped flask of total

$\pi H R$

3

volume The flow out at the bottom of the flask is proportional to the height (H) of the tank as shown below. Derive the equation describing the dynamics of the system

$F_o \text{ m}^3/\text{min}$

R

r

H

h

$F = k\sqrt{h} \text{ (m}^3/\text{min)}$

1.2 Process Control

In all processes there arises the necessity of keeping flows, pressures, temperatures, and composition etc. within certain limits for reasons of safety and or specification. In the process industries automatic control is highly desirable, as manual operation would necessitate continuous monitoring of the controlled variables by human operators, and the efficiency of observation of the operator would inevitably fall off with time.

The control of a process is most often accomplished by measuring the variable it is required to control (the control variable), comparing the measured values with the value desired to maintain the controlled variable (the set point), and adjusting some further variable (the manipulated variable) which has a direct effect on the controlled variable.

In order to design a control system to operate automatically and

efficiently, it is necessary to obtain both the steady state and the dynamic (unsteady state) relationships between the particular variable involved. The feedback and feed forward control systems are usually used. However, because of the development of the microprocessor the feedback system had gained wider application.

1.2.1 Block Diagram of a Feedback Control System

The blocks representing the various components of a control system are connected in a fashion which characterise their functional relationship within the system. This is represented below.

6

Disturbance

Actuating Manipulated Process

r

+ Control Control Output

c

signal

Set point Error c Element, g 1Variable, m g2

Feedback

Element, h

Feedback path

Figure 1.4 The Primary Feedback Control Loop

Some definitions applied to the primary feedback control loop:

- a) Manipulated Variable: This is the input to the system (process) that can be changed in order to control the plant or process
- b) Disturbance: This is an undesirable input signal which affects the value of the controlled output and other variables which we have no control over.
- c) Negative Feedback: This is when the summing point is added, i.e. $e = r + b$
- d) The Control Element, g: This is also called the controller. They are the components required to generate the appropriate control signal, m applied to the process.
- e) The Feedback Elements, h: This is the component required to established the functional relationship between the primary feedback signal, b, and the controller output, c
- f) The Reference Input, r: This is an external signal applied to a feedback control system in order to command a specified action of the process. It often represents ideal process output behaviour.
- g) Transducer: This is a device which converts one energy form into

another, e.g. potentiometer which converts mechanical position into an electric voltage.

The primary feedback control loop shown in Figure 1.4 can be represented in a canonical form as given below.

7

$r + GG c$

b

H

Figure 1.5 The Primary Feedback Control Loop

The following definitions refer to this block diagram

G = Direction transfer function

GH = loop transfer function – open loop

c/r = Primary feedback ratio

The following equations hold (-) sign refers to the positive feedback system and the (+) sign refers to the negative feedback

$c G$

$r 1 \pm GH$ 1.13

$c 1$

$r 1 \pm GH$ 1.14

$b GH$

$r 1 \pm GH$ 1.15

$1 \pm GH$ is the characteristic equation

The block diagram of a practical feedback control system can be quite complicated. However, it can be systematically reduced into a canonical form. The manipulation that effects the reduction are referred to as block diagram algebra and are restricted to systems which can be described by linear equations with constant coefficients

1.2.2 The Transfer Function

Generally, a piece of chemical plant can be represented by differential equations. When these are transformed we obtain a function representing the plant known as the transfer function. This function normally determines the ways in which the system reaches its new equilibrium position.

The transfer function is defined as the response of a control system to a disturbance of a unit impulse from steady state conditions. The transfer function describing a path of a complete control system is usually inserted into a block diagram that can be used to depict the action of the variable involved in automatic control theory.

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Laplace y transform of output

Transfer function =

Laplace y transform of input 1.16

$$X(s) G(s) Y(s)$$

$$Y(s)$$

$$G(s) =$$

$$X(s) \quad 1.17$$

$$Y(s) = G(s) \cdot X(s) \quad 1.18$$

$G(s)$ = Transfer function

The advantage of defining a transfer function in terms of Laplace transform of input and output is that the differential equations developed to describe the unsteady state behaviour of the system are reduced to simple algebraic relationships which can be solved using normal algebraic laws. Furthermore, the output (response) of the system to a variety of inputs may be obtained without classical integration.

1.2.3 Block Diagram Algebra

In order to employ the same signal or variable as an input to more than one block or summing point (cycle), a take-off point is used. This permits the signal to proceed unaltered along several different paths to several destinations e.g.

x

x x

Take off-point x

x G y

y

Take off-point

1.2.3.1 Block in Series

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$$X_1(s) X_2(s)$$

$$X(s) G_1(s) G_2(s) G_3(s) Y(s)$$

$$X(s)$$

$$G_1(s) =$$

$$X_1(s) \quad 1.19$$

$$X(s)$$

$$G_2(s) = 1$$

$$X_2(s) \quad 1.20$$

$$Y(s)$$

$$G_3(s) = \frac{1}{s^2 + 1.21}$$

However, to obtain $G(s)$ for blocks in series, we multiply all the transfer functions

$$G(s) = G_1(s) \cdot G_2(s) \cdot G_3(s) = \frac{1}{s^2 + 1} \cdot \frac{1}{s^2 + 1.21} = \frac{1}{s^4 + 2.21s^2 + 1.21} \quad 1.22$$

1.2.3.2 Block in Parallel

$$\begin{aligned} & y_1 \\ & G_1 \\ & y_2 \\ & x \quad G_2 \quad y \\ & y_3 \\ & G_3 \end{aligned}$$

Input x is applied to all blocks

$$\begin{aligned} & y \\ & G_1 = 1 \\ & x \quad 1.23 \end{aligned}$$

$$\begin{aligned} & y \\ & G_2 = 2 \\ & x \quad 1.24 \end{aligned}$$

$$\begin{aligned} & y \\ & G_3 = 3 \\ & x \quad 1.25 \end{aligned}$$

$$\begin{aligned} & 10 \\ & y \quad y \quad y \quad y + y + y \\ & G = G_1 + G_2 + G_3 = 1 + 2 + 3 = 6 \\ & x \quad x \quad x \quad x \quad 1.26 \end{aligned}$$

That is, the resultant transfer function of blocks in parallel is the summation of the individual transfer functions

1.3.2.3 Junction of Signals

$$\begin{aligned} & x_3 \\ & x_1 \quad y \quad x_1 \quad y \\ & x_2 \quad x_2 \\ & y = x_1 + x_2 \quad y = x_1 + x_2 + x_3 \end{aligned}$$

1.2.4 Reduction of Complicated Blocks

Complicated block diagrams are made up of loops and multiple inputs. The following are the steps involved in reducing a complicated block

diagram

Step 1

Combine all cascade blocks

$$x \rightarrow G_1 \rightarrow G_2 \rightarrow y \rightarrow x \rightarrow G_1 G_2 \rightarrow y$$

2

1

y

$$= G_1 G_2$$

$$x \therefore y = (G_1 G_2) x$$

1.27

Step 2

Continue all parallel blocks

11

$$x \rightarrow G_1 \rightarrow y \rightarrow x \rightarrow G_1 + G_2 \rightarrow y$$

G2

y

$$= G_1 + G_2$$

$$x \therefore y = (G_1 + G_2) x$$

1.28

Step 3

Represent the block in a canonical and determine the transfer function

G3

G4

$$x \rightarrow G_1 \rightarrow G_2 \rightarrow y \rightarrow x \rightarrow G_1 G_2 \rightarrow (G_3 + G_4) \rightarrow y$$

1

y

$$= G_1 G_2 - (G_3 + G_4) \therefore y = (G_1 G_2 - (G_3 + G_4)) x$$

x 1.29

2.0 Review of Laplace Transform

Standard transforms:

$$F(t) \rightarrow F(s) = \mathcal{L}[f(t)]$$

$$a \rightarrow a/s$$

1

$$at \rightarrow s - a$$

e

a

$$s + a^2$$

2

$\sin at$

12

s

$s + a^2$

2

$\cos at$

a

$s - a^2$

2

$\sinh at$

s

$s - a^2$

2

$\cosh at$

n!

$n s^n + 1$ (n = positive integer)

t

t^{n-1}

$(n-1)! s^n$

2.1 The Shift Theorems

Theorem 1:

If $\mathcal{L}\{F(t)\} = f(s)$, then $\mathcal{L}\{e^{-at}F(t)\} = f(s + a)$

Theorem II; Multiplying by t

d

$\{f(s)\}$

If $\mathcal{L}\{F(t)\} = f(s)$, then $\mathcal{L}\{tF(t)\} = -\frac{df}{ds}$

Theorem III: Dividing by t

$F(t)$

t

$=f(s)$ provided $\lim_{t \rightarrow 0} \frac{F(t)}{t}$ exists

$F(t)$

t

exists

If $\mathcal{L}\{F(t)\} = f(s)$, then

Examples

Determine the Laplace transform of the following functions

1) $\sin 3t$

$$2) t^3 - 3t + 4$$

$$13$$

$$e^{3t} - 1$$

$$3) t$$

$$4) e^{4t}$$

$$5) e^{3t} \cos 4t$$

$$6) t^2 \sin t$$

Solutions

$$3$$

$$2$$

$$1) \sin 3t = s - 9$$

$$2 \ 3 \ 4 \ 1 \ 1$$

$$-2 + = 3(2 - 3s + 4s^2) = 3(4s^2 - 3s + 2)$$

$$2) t^2 - 3t + 4 = s s s s$$

$$3$$

$$s$$

$$e^{3t} - 1$$

$$3) t$$

first check if the limit exists

$$e^{3t} - 1 \quad 3 \quad e^{3t}$$

lim

$$t \rightarrow 3$$

=0, applying L'Hopital rule lim

$$t \rightarrow 0 \quad t \rightarrow 0$$

The limit exists, then

$$(3e^{3t} - 1) = 3 - 1$$

$$s - a s$$

$$\int$$

$$[]$$

$$1$$

$$(t)(s - a s) - \ln(s - 3) - \ln s$$

$$3t \infty$$

$$3e^{-1} \ 3 \ 1$$

$$= \int -ds = 3$$

$$s$$

$$3s$$

$$\int$$

$$= -[\ln(s - 3) - \ln s] = \ln$$

$$[]s$$

$$s - 3$$

$$4) e^{4t} = s - 4$$

$$1$$

$$5) e^{3t} \cos 4t \text{ (applying theorem I)}$$

$$s$$

$$(\cos 4t) = 2$$

$$s + 16$$

$$\int$$

$$14$$

$$(e^{3t} \cos 4t) = 2s - 3 = 2$$

$$s - 3$$

$$s - 6s + 9 + 16s - 6s + 25$$

$$\int$$

$$6) t^2 \sin t \text{ (theorem II)}$$

$$1$$

$$2$$

$$\int(\sin t) = s - 1$$

$$\int(t \sin t) =$$

$$d$$

$$2$$

$$1$$

$$ds s - 1$$

$$(0 - 2s)$$

$$2$$

$$2s$$

$$(s - 1)(s + 1)$$

$$2 \ 2 \ 2$$

$$(s + 1)^2 - 2s(2(s + 1)^2 s)$$

$$2$$

$$()$$

$$2 \ 2$$

$$d \ 2s$$

$$- = -$$

$$ds (s^2 + 1)^2 (s^2 + 1)$$

$$4$$

$$\int(t^2 \sin t) =$$

$$2$$

$$-2(s^2 + 1) - 8s^2(s^2 + 1)$$

$$[\ 2$$

$$-(s^2 + 1)^2 (s^2 + 1) - 8s^2] = 6s^2 + 2$$

$$4 \ 3 \ 3$$

$$(s^2 + 1)(s^2 + 1)(s^2 + 1)(s^2 + 1)$$

2.2 Inverse Transform

Inverse transform is the reverse process of the Laplace transform i.e.

finding the function of t to which the Laplace belongs. It is written as $\mathcal{L}^{-1}$.

Note that this symbol does not indicate reciprocal e.g.

$$1$$

$$-1 \ s^2 - 1$$

$$(i) \ \mathcal{L} = e^{2t}$$

$$2s$$

$$= 2 \cos 5t$$

$$-1 \ s^2 + 25$$

$$(ii) \ \mathcal{L}$$

$$12$$

$$2$$

$$= 4 \sinh 3t$$

$$(iii) \ \mathcal{L}^{-1} \ s - 9$$

Therefore given a transform, we can write down the corresponding function in t provided we can recognise it from our table of transforms.

Let us consider

$$3s + 1$$

$$\mathcal{L}^{-1} \ s - s - 6$$

$$2)$$

This can only be transformed if we split it into two simpler functions such as

$$1 \ 2$$

$$s + 2 \ s - 3$$

$$15$$

$$1 \ 2$$

The two simpler fractions $s + 2$ and $s - 3$ are called the partial fractions of

$$3s + 1$$

$$2$$

$$s - s - 6$$

2.3 Rules of Partial Fraction Resolutions

1. The numerator must be of lower degree than the denominator. If it is not, then we divide out

2. Factorise the denominator into its prime factors. These determine the shape of the partial fractions

A

3. A linear factor $(s + a)$ gives a partial fraction $\frac{A}{s + a}$ where A is a

constant to be determined

A B

2

4. A repeated factor $(s + a)^2$ gives $s + a + (s + a)$

A B C

2 3

5. Similarly, $(s + a)^3 = s + a + (s + a) + (s + a)$

P s + Q

2

6. A quadratic factor $(s^2 + ps + q) = s + ps + q$

P s + Q R s + T

7. Repeated quadratic factors $(s^2 + ps + q)^2 =$

$s + ps + q (s + ps + q)^2 =$

2

2

Examples

Express the following functions in partial fractions and hence determine their inverse transforms

$5s + 1 \over s^2 - 15s + 41$ $4s^2 - 5s + 6$

2

(1) $s + s + 12$ (2) $(s + 2)(s - 3)^2$ (3) $(s + 1)(s^2 + 4)$

Solutions

$5s + 1 \over s^2 - 15s + 41$

(1) $s + s + 12 = (s + 3)(s - 4)$

2

Resolving into partial fractions yields

16

$5s + 1 \over s^2 - 15s + 41 = A \over s + 3 + B \over s - 4$

= +

$(s + 3)(s - 4) \over s + 3 s - 4$

$5s + 1 \equiv A(s - 4) + B(s + 3)$

If $s = 4$

$21 = 7B$ i.e. $B = 3$

If $s = 3$

$-14 = -7A$ i.e. $A = 2$

Therefore

$5s + 1 \over s^2 - 15s + 41$

$\equiv +$

$(s + 3)(s - 4) \over s + 3 s - 4$

$$(5s + 1)$$

$$) () ()^2 3$$

$$-1 -1$$

$$\equiv f + f$$

$$f^{-1}(s+3)(s-4)$$

$$s+3 \quad s-4 = 2e^{-3t} + 3e^{4t}$$

Work through the other two examples

2.4 Poles and Zeros

$$\Phi(s)$$

Generally, Laplace transform has the form $f(s) = \frac{\theta(s)}{\Phi(s)}$ with the

$$\Phi(s)$$

denominator factoring into $(s-a)(s-b)(s-c)\dots$

Poles: The values $a, b, c, \dots$ that make the denominator zero and hence $f(s)$ infinite are called the poles of $f(s)$. If there are no repeated factors, the poles are simple poles. If there are repeated factors the poles are multiple poles

Zeros: The values of s that make the numerator zero and hence $f(s)$ zero

$$s+4 \quad 3$$

$$2 -$$

are called zeros of $f(s)$, e.g. $(s+2)(2s+3)$ has a simple pole at $s = 2$ and a double pole at $s = -2$ and a zero at $s = -4$

2.5 Solution of Differential Equations by Laplace

To solve a differential equation by Laplace, we follow four distinct steps

1) Re-arrange the equation in terms of Laplace transforms

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2) Insert the given initial conditions

3) Re-arrange the equations algebraically to give the transform of the solution

4) Determine the inverse transform to obtain the particular solution

Let the Laplace transform of x be denoted by X

$$X = f(X) = f\{F(t)\} = f(s)$$

i.e. (2.1)

Then using the 'dot' notation for differential coefficient, we have

$$f(\dot{x}) = X$$

(2.2)

$$f(\ddot{x}) = sX - x_0 \quad (2.3)$$

$$f(\ddot{\ddot{x}}) = s^2 X - sx_0 - \dot{x}_1 \quad (2.4)$$

$$f(\ddot{\ddot{\ddot{x}}}) = s^3 X - s^2 x_0 - s\dot{x}_1 - \ddot{x}_2 \quad (2.5)$$

Examples

(1) Solve the equation

$\frac{dx}{dt}$

$$+2x = 10e^{3t}$$

given that $t = 0, x = 6$

Converting into Laplace domain

10

$$(sX - x_0) + 2X = \frac{10}{s-3}$$

$$, x_0 = 6$$

10

$$sX + 2X - 6 = \frac{10}{s-3} \therefore (s+2)X = \frac{10}{s-3} + 6$$

$$s-3 \quad s-3$$

$$6s-8 \quad 6s-8$$

$$(s+2)X = \frac{6s-8}{(s-3)(s+2)}$$

$$s-3 \quad (s-3)(s+2)$$

Transforming into partial fraction

$$6s-8 = \frac{A}{s-3} + \frac{B}{s+2}$$

$$\equiv +$$

$$(s-3)(s+2) \quad (s-3)(s+2)$$

$$6s-8 \equiv A(s+2) + B(s-3)$$

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If $s = -2; B = 4$ and if $s = 3, A = 2$

$$(6s-8)$$

$$)(s-3)(s+2)$$

$$-1 \quad -1 \quad -1$$

$$\int \frac{2}{s-3} + \frac{4}{s+2}$$

$$s-3$$

$$s+2$$

$$x = 2e^{3t} + 4e^{-2t}$$

$$\frac{d^2x}{dt^2}$$

2

$$-4x = 24 \cos 2t$$

(2) Solve $\frac{dx}{dt}$

$\frac{dx}{dt}$

$$= 4$$

Given that at $t = 0, x = 3$, and $\frac{dx}{dt}$

Transforming yields

24

$$(s^2X - sx_0 - x_1) - 4X = \frac{24}{s^2+4}$$

Initial conditions; $x_0 = 3$, $x_1 = 4$

Substituting these values

$$24s + 24$$

$$s^2 x - 3x_0 - 4 - 4x = 2 \therefore (s^2 + 4)x = 2 + 3s + 4$$

$$s + 4s + 4$$

$$3s^3 + 4s^2 + 36s + 16$$

$$\Rightarrow x =$$

$$(s^2 + 4)(s - 2)(s + 2)$$

Resolving into partial fraction

$$3s^3 + 4s^2 + 36s + 16 = \frac{A}{s - 2} + \frac{B}{s + 2} + \frac{Cs + D}{s^2 + 4}$$

$$= + +$$

$$(s^2 + 4)(s - 2)(s + 2) \left[\frac{A}{s - 2} + \frac{B}{s + 2} + \frac{Cs + D}{s^2 + 4} \right]$$

$$= 3s^3 + 4s^2 + 36s + 16 = A(s - 2)(s^2 + 4) + B(s + 2)(s^2 + 4) + (Cs + D)(s - 2)(s + 2)$$

$$\text{If } s = 2, B = 4; \text{ if } s = -2, A = 2$$

Equating coefficients of s^3 and constant terms, $C = -3$, and $D = 0$

Giving

$$3s^3 + 4s^2 + 36s + 16 = \frac{2}{s - 2} + \frac{4}{s + 2} + \frac{-3s}{s^2 + 4}$$

$$x = \frac{2}{s - 2} + \frac{4}{s + 2} - \frac{3s}{s^2 + 4}$$

$$(s^2 + 4)(s - 2)(s + 2) \left[\frac{2}{s - 2} + \frac{4}{s + 2} - \frac{3s}{s^2 + 4} \right]$$

Then

$$19$$

$$-1$$

$$\int x = 2e^{-2t} + 4e^{2t} - 3 \cos 2t$$

3.0 Dynamic behaviour of processes and process plant control

3.1 First Order Systems

3.1.1 Concentration Response of CSTR: System Without Chemical Reaction

A well stirred tank generally responds as a first order system to changes in input concentration. Consider a CSTR with no chemical reaction. The dynamic equation is:

$$V \frac{dC_A}{dt} = FC_{Ao} - FC_A$$

$$\text{or } F(C_{Ao} - C_A) = V \frac{dC_A}{dt}$$

$$(3.1)$$

$$F \frac{dC_A}{dt}$$

$$V \frac{dC_A}{dt}$$

$$V \frac{dC_A}{dt}$$

$$(C_{Ao} - C_A) =$$

dt (3.2)

V

= τ

Let F

Equation (3.2) can be re-written as

$$1 \frac{1}{\tau} \frac{dC_A}{dt} = 1 \frac{1}{\tau} (C_{Ao} - C_A)$$

$$\Rightarrow C_{Ao} - C_A = \tau \frac{dC_A}{dt} \quad (3.3)$$

$$\tau \frac{dC_A}{dt} = C_{Ao} - C_A$$

Transforming equation (3.3)

$$1 \frac{1}{\tau} \frac{dC_A}{dt} = 1 \frac{1}{\tau} (C_{Ao} - C_A)$$

$$sC_A + C_A = C_{Ao} \Rightarrow s + C_A = C_{Ao}$$

$$\tau \frac{dC_A}{dt} + C_A = C_{Ao} \quad (3.4)$$

$$1$$

$$C_A = C_{Ao}$$

$$\tau \frac{dC_A}{dt} = C_{Ao} - C_A$$

$$\therefore C_A = \frac{C_{Ao}}{1 + s\tau}$$

$$1 \frac{1}{s\tau + 1}$$

$$s + \frac{1}{\tau}$$

$$\tau \frac{dC_A}{dt} + C_A = C_{Ao} \quad (3.5)$$

$$C_A = \frac{C_{Ao}}{1 + s\tau}$$

$$\therefore C_A = \frac{C_{Ao}}{1 + s\tau}$$

$$C_A = \frac{C_{Ao}}{1 + s\tau} \quad (3.6)$$

The following is the representation of the CSTR

F

C_{Ao}

20

$C_{Ao} - C_A$

$\tau \frac{dC_A}{dt} + C_A = C_{Ao}$

3.1.2 Concentration Response of CSTR: System with Chemical Reaction

If the CSTR is a chemical reactor, the quantity reacting per unit time is an additional output term in the material balance for the reactant. For a first order reaction, the reaction rate is proportional to the concentration of reactant in the tank, which is the same as the exit concentration for a CSTR. The quantity is:

$$r \text{ (kg h}^{-1}\text{)} = k \text{ (h}^{-1}\text{)} \cdot C_A \text{ (kg m}^{-3}\text{)} \cdot V \text{ (m}^3\text{)}$$

$$(3.7)$$

$$r = k C_A V \quad (3.8)$$

The material balance for the reactant is

dC_A

$F C_{Ao} - F C_A - k C_A V = V \frac{dC_A}{dt}$

$\frac{dC_A}{dt}$

(3.9)

V

$= \tau$

Let F

Equation (3.9) can be re-written as

$V \frac{dC_A}{dt} + C_A + k C_A = C_{Ao}$ or

$\tau \frac{dC_A}{dt} + C_A + k C_A = C_{Ao}$

1

$+ C_A + k C_A = C_{Ao}$ or

$\tau \frac{dC_A}{dt} + C_A + k C_A = C_{Ao}$

1

$+ C_A + k C_A = C_{Ao}$

$\tau \frac{dC_A}{dt} + C_A + k C_A = C_{Ao}$ (3.10)

Transforming equation (3.10)

$s C_A + (1 + k) C_A = 1 C_{Ao}$

$A C_{Ao} (1 + k)$

$) 1$

$\Rightarrow s + 1 + k C_A = C_{Ao}$

$\tau \frac{dC_A}{dt} + C_A + k C_A = C_{Ao}$ (2.11)

1

$C_A \tau 1$

$\therefore =$

$C_{Ao} 1 + \tau s + 1 + k$

$s + 1 + k$

τ (2.12)

Putting equation (2.12) in the form of equation (3.6), we have

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$C_A K$

$\therefore =$

$C_{Ao} \tau s + 1$ (2.13)

Where

F

$K =$

$kV + F$ System gain

V

$V F$

$$\tau = \frac{V}{F}$$

$$kV + F kV$$

$$+1$$

F Time constant

The effect of the chemical reaction is to make the time constant of the system less than the hold-up time. Equation (4.13) shows that as the rate constant becomes very large, the time constant approaches $1/K$. a fast reaction also means a low value of K - the steady state gain, and this may help to explain effect of K on the time constant and hold-up time. After a step change in C_{Ao} , the initial rate of change of C_A is initially zero. With a large K , the eventual change in C_A is small and so less time is required to accomplish a given fraction of the change.

3.1.3 Reactions Other Than First Order

If the reaction is zero order, τ is again V/F (the hold-up time), since change in concentration do not change reaction rates. For reactions which are second or fractional order, the exact response cannot be obtained by using transfer functions, since the differential equations are not linear. However, the response to changes can be predicted quite well by using a linear approximation of the rate expression as shown below.

For second order reaction:

$$r = C_2 A^2 V \quad (2.14)$$

$$r \approx r +$$

$$\frac{\partial r}{\partial C_A}$$

$$(C_A - C_A)$$

$$\frac{\partial C_A}{\partial C_A} C_A A$$

$$(3.15)$$

From equation (3.14)

$$\frac{\partial r}{\partial C_A}$$

$$= 2 k V \{ C_A \}$$

$$\frac{\partial C_A}{\partial C_A} A \quad (3.16)$$

Substituting equation (3.16) into (3.15) yields

$$22$$

$$r = r + 2 k V \{ C_A (C_A - C_A) \}$$

$$(3.17)$$

or

$$r = r + V C_A (2 k C_A)$$

$$\text{if } C_A = C_A - C_A \quad (3.18)$$

Equation (3.18) can be used in a material balance similar to equation (3.9) or the transfer function can be written directly from equation (3.13)

by substituting the term

$2 k C_A$ for the first order rate constant k i.e.

$C_A K$

$C_{Ao} \tau_s + 1$ (3.19)

Where

F

$K =$

$2 C_A k V + F$ System gain

V

$V F$

$\tau =$

$2 C_A k V + F$ $2 C_A k V$

$+ 1$

F Time constant

In equation (3.19), C_{Ao} , C_A are transformed variables representing deviations from normal input and output concentrations.

3.1.4 Temperature Response of CSTR

Consider a jacketed tank used to preheat a process stream shown below

F

θ_1

steam F

θ_s θ_2

K_s

θ_s θ_s θ_1

condensate K_1 $+$ $+$

θ_1 θ_2

θ_1 θ_s θ_1

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Assuming the fluid temperature in the tank is uniform, then the system is first order.

Material balance:

$d\theta_2$

$F\rho C_P \theta_1 - F\rho C_P \theta_2 + UA (\theta_s - \theta_2) = MCP$

dt

(3.20)

$MCP d\theta_2 F\rho C_P UA$

$+ \theta_2 = \theta_1 + \theta$

$F\rho C_P + UA dt F\rho C_P + UA F\rho C_P + UA s$ (3.21)

$d\theta_2$

$$\frac{d\theta_2}{dt} = K_1 \theta_1 + K_2 \theta_s$$

dt

(3.22)

$$\therefore (\tau_s + 1) \theta_2 = K_1 \theta_1 + K_2 \theta_s$$

(3.23)

The transfer functions for the tank are obtained by considering the initial values of θ_1 , θ_2 and θ_s to be zero which means that these variables stand for deviations from the average values.

If the response of the system to changes in feed temperature is desired, θ_s in equation (3.23) is set to zero, and the standard solution to step and ramp or other changes in θ_1 is applied. For changes in steam temperature, feed temperature is considered constant and θ_1 term dropped. If the steam temperature is adjusted to control θ_2 , changes in θ_1 would be load changes and the transfer function for θ_1 would be generally shown as a separate branch of the block diagram. However,

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since the time constant is the same for changes in either steam temperature or feed temperature, the load signal can be introduced ahead of the major block if multiplied by the gain ratio as shown below.

K_1

θ_1

$\frac{1}{1 + \tau_1 s}$

$\theta_C + \theta_s K_2$

$K_C \theta_2$

$\frac{1}{1 + \tau_2 s}$

K_1

θ_1

K_2

$+ + K_2$

$\theta_C K_C \theta_2$

$+ \frac{1}{1 + \tau_2 s}$

3.1.5 Response to Changes in Flow

The rate of flow to the tank might be a variable, and if we consider only small changes in flow, the effects of these changes on the time constant and hold-up time can be neglected. Here we have to use separate symbols for the average flow F and the deviation ΔF and the symbols θ_1

θ

and 2 represent average values of the temperature on a common scale. It can be shown that the change in flow rate just adds one extra (input – output) term to equation (3.20)

$d\theta_2$

$$F \rho C_p (\theta_1 - \theta_2) + UA (\theta_s - \theta_2) + \Delta F \rho C_p (\theta_1 - \theta_2) = MCP \frac{d\theta_2}{dt}$$

dt

(3.24)

25

$$= (\tau_s + 1) \theta_2 = K_1 \theta_1 + K_2 \theta_s + K_3 \Delta F$$

(3.25)

Where τ , K_1 , and K_2 are the same as in equations (3.21) and (3.22), while

$$\rho C_p (\theta_1 - \theta_2)$$

$K_3 =$

$$\frac{-\Delta F \rho C_p}{\rho C_p + UA}$$

(3.26)

The value of K_3 is negative since an increase in flow rate decreases the output temperature.

For the CSTR, the time constant for the response of the tank temperature is the same for changes in feed temperature. Steam temperature or flow rate, but the steady state gains are different for each input variables.

Recognising that the time constant is the same for several inputs saves time in deriving transfer functions for control systems. Once the time constant is determined, it is necessary only to get the steady state gain for different inputs, and these gains are often obtained from routine steady state calculations rather than from the complete differential equations.

For CSTR, the time constant for temperature changes is $M C_p / (F \rho C_p + UA)$. Note that the time constant approaches M/F or V/F - the hold-up time when the flow rate is high. At low flow rates the time constant approaches $M C_p / UA$. It is important to remember that the time constant

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for temperature changes generally differs from that for level changes, though these time constants are often the same in magnitude as the hold-up time

4.0 Open Loop Response of Simple Systems

4.1 Step Input

The transform of a first order system is

C_{Ao}

$C_A(s) =$

$\frac{\tau s + 1}{s}$

(4.1)

If $C_{Ao} = b$ (a step change)

$-1/b$

$\Rightarrow \int C_{Ao} =$

s

(4.2)

$-1/b + 1/b$

$\therefore \int C_A(s) =$

$\frac{s\tau + 1}{s(\tau s + 1)}$

(4.3)

By partial fraction transformation

$b/A + B$

$s(\tau s + 1) = s\tau + 1$

(4.4)

$\Rightarrow b = A(\tau s + 1) + Bs$

If $s = 0$: $A = b$ and if $(\tau s + 1) = 0$, then

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$S = -1/\tau$, i.e. $B = b\tau$

Substituting these values into equation (3.3), we have

$b/b\tau + b/b$

$C_A(s) = \frac{b}{s} + \frac{b\tau}{s\tau + 1}$

$\frac{b}{s} + \frac{b\tau}{s\tau + 1}$

τ

(4.5)

()

$-1 - \tau - \tau$

$\therefore \int C_A(s) = b - b\tau e^{-t/\tau}$

$\tau \tau$

(4.6)

The general solution for the response to a step input is given in terms of functional response, y , i.e.

()

$-t$

$C_A(s) = b(1 - e^{-t/\tau})$

(4.7)

$C_A - C_A(t)$

$y = 1 - e^{-t/\tau}$

$b C_A$

s. s (4.8)

Equation (4.8) gives the response of any first order process to a step input and it is used so often that it should be memorised. When $t/\tau = 1$, the functional response is $1 - e^{-1}$ or 0.632. This time constant can be defined as the time for system's output to reach 63.2% of its final value after a step change in input. The terms time constant, first order lag, first order elements, and exponential lag, all refers to portions of a system which have first order behaviour.

28

1.0

t

0.632

0

1.

0

y

A more exact method for determining the time constant for true first order systems is to use the slope of a semi-log plot of the uncompleted response $1 - y$ or $b(1 - y)$ against time. A semi-log plot shows any deviation from first order behaviour more clearly than an arithmetic plot. A third way of determining the time constant is from the initial slope of the response curve which is equal to the steady state situation over the time constant

4.2 Ramp Input

For a ramp input $C_{Ao} = bt$, therefore

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$\int_0^t C_{Ao} dt = b/s^2$

(4.9)

$b \cdot 1/b$

$C_A(s) = \frac{2}{s^2} = \frac{2}{s(s + 1/\tau)}$

$s \tau s + 1 \quad s(\tau s + 1)$

(4.10)

Partial fraction transformation gives

$$b A B C$$

$$= + 2 +$$

$$s (\tau s + 1) s s \tau s + 1$$

$$2$$

$$(4.11)$$

$$b = A(\tau s + 1)s + B(\tau s + 1) + Cs^2$$

$$(4.12)$$

$$\text{if } s = 0, B = b; \text{ and if } s = -1/\tau, C = b\tau^2$$

equating coefficient of s^2

$$b\tau^2$$

$$A = -b\tau$$

$$\tau$$

Substituting these values into equation (4.10)

$$b b\tau^2 b\tau b b\tau b\tau$$

$$C A (s) = \frac{2}{s^2} + \frac{-b\tau}{s} + \frac{b\tau^2}{s + 1/\tau}$$

$$s \tau s + 1 s s s + \tau s$$

$$(4.13)$$

$$() () ()$$

$$-t$$

$$-1 b b\tau b\tau t - t \tau$$

$$\therefore C A (s) = \int \frac{2}{s^2} + \frac{-b\tau}{s} + \frac{b\tau^2}{s + 1/\tau} = b t + \tau e^{-t/\tau}$$

$$- \tau = b\tau + e^{-1}$$

$$s^2 s + 1 \tau s \tau$$

$$(4.14)$$

$$30$$

The response of a first order system to a ramp input is given by equation

(4.14). after the transient term $e^{-t/\tau}$ becomes negligible and the output is

$b\tau$ less than the input or the output lags the input by a time equals the

time constant, i.e.

$$CA = bt - b\tau = b(t - \tau)$$

$$(4.15)$$

$$CA$$

$$CA_0$$

$$CA_1$$

$$b\tau$$

$$t$$

Examples

1, Gases A and B are fed continuously to a tank with a volume of 30 m³. The normal flow rate of these gases measured at tank conditions of 40 bar and 80°F are $F_A = 40 \text{ m}^3/\text{min}$ and $F_B = 10 \text{ m}^3/\text{min}$. If the flow of B is suddenly increased to 12 m³/min, when does the concentration of B reach 90% of the new steady state value?

suddenly increased to 12 m³/min, when does the concentration of B reach 90% of the new steady state value?

Solution

The time constant for small changes in flow rate is the normal hold-up time, i.e.

$$V = 30$$

$$\tau = \frac{V}{F} = \frac{30}{40+10} = 0.6 \text{ min}$$

$$F = 40+10$$

$$y = 1 - e^{-t/\tau} = 0.9$$

therefore,

$$t/\tau = 2.3 \text{ and } t = 1.4 \text{ mins}$$

2. A first order reaction is carried out in a CSTR with a hold-up time of 1.6 hr and a rate constant of 2 h⁻¹. Show the effect of a sudden change in feed concentration from 0.50 to 0.48 mol/m³

Solution

Initial conditions are

$$F(C_1 - C_2) = k C_2 V$$

where C_1 and C_2 are the normal inlet and exit concentrations

$$32$$

$$C_2 = 1$$

$$= 0.238 \therefore C_2 = 0.119$$

$$C_1 = kV$$

$$1 + F = 1 + 3.2$$

The transfer function is

$$C_2 = K \frac{1}{\tau s + 1}$$

$$= \text{and } K = kV$$

$$= 0.238$$

$$C_1 = \frac{kV}{\tau s + 1} = \frac{1}{1 + 4.2s}$$

$$V$$

$$F = 1.6$$

$$\tau = kV$$

$$= 0.381 \text{ hr}$$

$$1 + 4.2s$$

F

The transient response to a step input is $y = 1 - e^{-t/\tau}$. The steady state change in concentration is $0.238(-0.02) = -0.00476$

C2 C2 + C2

 t/τ t (hr) y

0 0 0 0 0.119

1 0.38 0.63 -0.0030 0.116

2 0.76 0.86 -0.0041 0.1149

3 1.14 0.95 -0.0045 0.1145

 ∞ ∞

1.0 -0.0048 0.1142

5.0 Second Order Systems: The Manometer

If the mechanical parts of a system or the fluid in a system are subject to accelerations. The dynamic behaviour is described by second order equations. A mass suspended from a spring is the classic example of a second order system, but a manometer is chosen as an example of more interest to chemical engineers. The basic equation is a force balance, and to simplify matters the frictional pressure drop is assumed proportional to the velocity and all the fluid is assumed to accelerate uniformly

33

h

L

The U - Tube Manometer

The mass balance from Figure (6.1) is

 $AL \rho \frac{d^2 h}{dt^2}$ $= A(P - \rho g h)$ $= A(P - \rho g h)$

g

gC

-RA

)dh

dt

(5.1)

Where

A = cross sectional area

 ρ = liquid density (density of gas above fluid is negligible)

P = applied pressure

R = frictional resistance

With laminar flow, the resistance is given by the Hagen-Poiseuille equation

$$\Delta P = \frac{32 \mu L Q}{\pi R^4} \quad \text{or } R = \frac{8 \mu L}{\pi R^4} \quad (5.2)$$

Substituting equation (5.2) into (4.1) yield

$$\frac{L}{\pi R^4} \frac{d^2 h}{dt^2} + \frac{8 \mu L}{\pi R^4} \frac{dh}{dt} = \frac{\rho g h}{\rho g} \quad (5.3)$$

Equation (5.3) may be written in standard form as

$$\frac{1}{\omega_n^2} \frac{d^2 h}{dt^2} + \frac{2 \xi}{\omega_n} \frac{dh}{dt} + h = h_i \quad (5.4)$$

Where ω_n = natural frequency, rad/sec

ξ = damping coefficient

The significance of ω_n and ξ becomes apparent after considering the solution to equation (5.4) for a step change in input pressure. With a damping coefficient less than 1, the output overshoots the final value and oscillates before coming to equilibrium. The system is said to be under-damped

For $\xi < 1.0$

$$h = h_i \left[1 - e^{-\xi \omega_n t} \sin(\omega_n \sqrt{1-\xi^2} t - \phi) \right] \quad (5.5)$$

$$\phi = \tan^{-1} \frac{1-\xi^2}{\xi} \quad (5.6)$$

35

With damping coefficient of zero, the response is an un-damped sine

wave of frequency ω_n and amplitude $2h_1$

With a damping coefficient of 1.0 the system ($\xi = 1.0$) is critically damped and comes to rest without over-shooting i.e. the response of first order system

For $\xi = 1.0$

$$h - \xi \omega_n t \\ = 1 - [1 - (\omega_n t)] e^{-\omega_n t} \quad (5.7)$$

For $\xi > 1.0$

The system is over-damped and comes to equilibrium slowly. When $\xi > 1$, the quadratic term can be factored and the solution is given in terms of two roots, i.e.

$$s^2 + 2\xi\omega_n s + \omega_n^2 = (\tau_a s + 1)(\tau_b s + 1) \quad (5.8)$$

$$h_1 = 1 + \tau_a e^{-t/\tau_a} - \tau_b e^{-t/\tau_b} \quad (5.9)$$

Experimental values of ω_n and ξ can easily be obtained from under-damped response curves of the type shown in the figure below. The

damping coefficient is found either from the decay ratio, which is the ratio of successive peak heights or from the maximum overshoot.

$$\text{Decay ratio} = e^{-2\pi\xi / \sqrt{1-\xi^2}} \quad (6.10)$$

$$\text{Maximum overshoot} = e^{-\pi\xi / \sqrt{1-\xi^2}} = \sqrt{\text{Decay ratio}}$$

Final value

(5.11)

The frequency of the damped oscillations is only slightly lower than the natural frequency for values of ξ between 0 and about 0.5

Frequency of damped oscillation $= \omega_n \sqrt{1 - \xi^2}$

(5.12)

1.0

overshoot overshoot

$\frac{b}{a}$ decay ratio $\frac{b}{a}$

finalvalue

a

time

Decay ratio b/a

1.8 ξ

1.6

1.4 $\xi = 0.1$

1.2

Fractional $\xi = 0.5$

response 1.0 $\xi = 1.0$

0.8

0.6 $\xi = 2.0$

0.4

37

0.2

1 2 3 4 5 6 7

$\omega_n t$

Fig 5.1 Step response of a second order system

5.1 Second Order ODE with Constant Coefficient

These systems can give an oscillatory or under-damped response.

Consider the second order ODE of the form

$\frac{d^2 x}{dt^2} + 2\tau_p \xi \frac{dx}{dt} + x = Q(t)$

(5.13)

First obtain the homogeneous solution, i.e $Q(t) = 0$

The transformed characteristic equation is

$\tau_p^2 s^2 + 2\tau_p \xi s + 1 = 0$

(5.14)

And by quadratic formula, we have

$$s = \frac{-2\tau_p \xi \pm \sqrt{4\tau_p^2 \xi^2 - 4\tau_p^2}}{2\tau_p^2}$$

$$(5.15)$$

The values of s that satisfy the characteristic equation are called eigen values of the system (or the roots of the characteristic equation)

$$x_h = C_1 e^{s_1 t} + C_2 e^{s_2 t}$$

$$(5.16)$$

$$s_1 = \frac{-\tau_p \xi \pm \sqrt{\tau_p^2 \xi^2 - 1}}{\tau_p^2}$$

$$(5.17)$$

$$s_2 = \frac{-\tau_p \xi \mp \sqrt{\tau_p^2 \xi^2 - 1}}{\tau_p^2}$$

$$(5.18)$$

The slope of the solution curve depends strongly on the values of the physical parameter, ξ - the damping coefficient. The following criteria applies

Case I: $\xi > 1$, the quantity inside the square root is positive, s_1 and s_2 will both be real and they will be different (i.e. distinct root). The system is over-damped

Case II: $\xi = 1$, the system is under-damped. The term inside the square roots is negative giving an imaginary root. The system becomes oscillatory and as long as ξ is positive (> 0), the exponential term in the homogeneous solution will decay to zero as time tends to infinity.

Therefore, the amplitude of the oscillations will decrease to zero. For examples consider the ODE

$$d^2 x + 5 dx + 6 x = 0$$

$$dt^2 dt$$

$$(5.19)$$

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Equation (5.19) can be written in the standard form as

$$\begin{aligned} & \left(\frac{1}{2} \frac{d^2 x}{dt^2} + \sqrt{6} \frac{dx}{dt} + 2 \right) \\ & \frac{1}{\sqrt{6}} \left(\frac{d^2 x}{dt^2} + 2\sqrt{6} \frac{dx}{dt} + x \right) = 0 \end{aligned} \quad (5.20)$$

Where

$$\tau_p = \left(\frac{1}{\sqrt{6}} \right) \text{ and } \xi = \left(\frac{2\sqrt{6}}{5} \right)$$

Then from the standard equation

$$\begin{aligned} s = & \frac{-2\tau_p \xi \pm \sqrt{4\tau_p^2 \xi^2 - 4\tau_p^2}}{2\tau_p^2} \\ & \xi \sqrt{\xi^2 - 1} \end{aligned} \quad (5.21)$$

we have

$$\begin{aligned} s = & - \left(\frac{1}{\sqrt{6}} \right) \left(\frac{\sqrt{24}}{5} \right) \\ & \pm \frac{2\sqrt{6} \sqrt{\left(\frac{2\sqrt{6}}{5} \right)^2 - 1}}{2 \left(\frac{1}{\sqrt{6}} \right)^2} \\ & = - \frac{5 \pm \sqrt{25 - 24}}{2 \left(\frac{1}{\sqrt{6}} \right)} \end{aligned} \quad (5.22)$$

$$\begin{aligned} s = & - \frac{5 \pm 1}{2 \sqrt{6}} \\ & = - \frac{5 \pm 1}{2 \sqrt{6}} \end{aligned} \quad (5.23)$$

Giving

$$\begin{aligned} s_1 = & - \frac{5 + 1}{2 \sqrt{6}} \\ & = -2 \end{aligned} \quad (5.24)$$

and

$$\begin{aligned} s_2 = & - \frac{5 - 1}{2 \sqrt{6}} \\ & = -3 \end{aligned} \quad (5.25)$$

The homogeneous solution is

40

$$x_h = C_1 e^{-2t} + C_2 e^{-3t}$$

(5.26)

The behaviour of the system depends strongly on the value of ξ , the damping coefficient. Assuming that the system above is forced with a unit step function, i.e

$$d^2 x / dt^2$$

$$+ 5 \frac{dx}{dt} + 6x = 1$$

$$dt^2$$

(5.27)

$$dx$$

$$x(t) = 0$$

$$dt$$

$$t = 0 \quad (5.28)$$

Let us find the particular solution by assuming that it is a constant (since the forcing function is a constant i.e. $x_p = C_3$). Substituting these conditions into equation (5.27) gives

$$0 + 5(0) + 6(C_3) = 1; \text{ i.e. } C_3 = 1/6$$

(5.29)

So the total solution is

1

$$x = x_h + x_p = C_1 e^{-2t} + C_2 e^{-3t} + \frac{1}{6}$$

6

(5.30)

which gives

1

$$x(t) = 0; \Rightarrow C_1 + C_2 + \frac{1}{6}$$

6

(5.31)

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$$dx$$

$$x(t) = 0 \Rightarrow -2C_1 - 3C_2$$

$$dt$$

$$t = 0 \quad (5.32)$$

Solving equations (5.31) and (5.32) simultaneously yield

1 1

$C_1 = -$; and $C_2 =$

$\frac{2}{3}$

The final solution is

$\frac{1}{3} \frac{1}{2} \frac{1}{6}$

$x(t) = e^{-3t} - e^{-2t} +$

$\frac{3}{2} \frac{2}{6}$

(5.33)

Again consider the ODE

$\frac{d^2 x}{dt^2} + x = 0$

$+ x = 0$

$\frac{d^2 x}{dt^2} + x = 0$

(5.34)

which can be written in the form

$\frac{d^2 x}{dt^2} + x = 0$

$+ 2(0.5) + x = 0$

$\frac{d^2 x}{dt^2} + x = 0$

(5.35)

i.e. $\tau_p = 1$, and $\xi = 0.5$

The characteristic equation is

$s^2 + s + 1 = 0$

(5.36)

or

$\frac{4}{2}$

$\xi = \frac{1}{2}$ and $\sqrt{3}$

$s = -\frac{1}{2} \pm \sqrt{1 - \xi^2} = -\frac{1}{2} \pm$

$\frac{\tau_p}{2} \pm \frac{\tau_p}{2}$

(4.37)

Therefore the homogeneous or complimentary solution is

$(\sqrt{3}t + 2C_1 \sin \sqrt{3}t$

$- st$

R

$x_h = C_1 e^{2}$

$2C \cos$

$2 \frac{2}{2}$

(5.38)

Where R and I are the real and imaginary parts of the complex number C .

The following cases arise with respect to equation (5.38)

Case I: $\xi = 0$, the system is under-damped, there is no decay of the sine

and cosine term and therefore, the system will oscillate for ever.

Case II: $\xi < 0$, the system is unstable. If ξ is negative, the exponential term increase without bound as time becomes large and the system is unstable. Thus, for

$$\xi < 0$$

$$s = -\xi \pm \sqrt{1 - \xi^2}$$

$$\tau_p \tau_p$$

$$(5.39)$$

If the real part of the root ($-\xi/\tau_p$) is negative, the term inside the square root is negative giving an imaginary number in the roots

$$\xi < 0$$

$$s = -\xi \pm \sqrt{1 - \xi^2} = -\xi \pm j\sqrt{1 - \xi^2}$$

$$\tau_p \tau_p \tau_p \tau_p$$

$$(5.40)$$

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The roots are now complex number with real and imaginary parts

$$\xi < 0$$

$$s_1 = -\xi + j\sqrt{1 - \xi^2}$$

$$\tau_p \tau_p$$

$$(5.33)$$

$$\xi < 0$$

$$s_2 = -\xi - j\sqrt{1 - \xi^2}$$

$$\tau_p \tau_p$$

$$(5.41)$$

The complimentary solution is

$$s_1, s_2$$

$$x_R = C_1 e^{s_1 t} + C_2 e^{s_2 t}$$

$$(5.42)$$

6.0 Control Instrumentation

The basic feedback control loop consists of a sensor to detect the process variable, a transmitter to detect and convert the sensor signal into an equivalent pressure and electrical signal, a controller that compares these process signals with a set point and produces an appropriate control signal and a control element that changes the manipulative variable. Usually, the final control element is an air operated control valve that opens and closes to change in the flow rate of the manipulated stream. The sensor, transmitter and control valve are physically located on the process equipment. The controller is usually located on a panel in the control room that is some distance away from the process equipment

Most equipment in chemical and petrochemical plants is pneumatic or electronic. Pneumatic devices are driven or powered by an instrument or stream that is supplied at about 20 – 25 psig. They usually accept and produce pneumatic input and output air signals of 3 – 15 psig. They are widely used because they are reliable, inexpensive and explosion proof. In a pneumatic system, the plastic or metal tubing that connects the transmitter to the controller and, the controller to the valve represents a dynamic lag in the control loop. When long lengths of tubing are used, these transmission lags can become significant and degrade control performance in fast loops.

6.1 Controllers

The controller mechanism may be considered as consisting of two sections, the comparator and the controller itself. The purpose of the first is to compare the measured and desired (set point) values of the controlled variable and to compute the difference between them as the

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error. The second section operates to alter the setting of the final control element (valve) in such a way as to minimise the error in the least possible time with minimum disturbance to the system. The controller action selected for the controller depends upon the dynamic behaviour of the other components in the control loop. There are three principal types of model of control action which are more generally employed

- Proportional Action (P)
- Integral or reset Action (I)
- Derivative or pre-act Action (D)

6.1.1 Proportional Action

A proportional feedback controller only changes the output signal in proportion to error signal

$$M(t) = M + KCe \quad (6.1)$$

where

$M(t)$ = controller output pressure that goes to control valve (Psig)

M = controller output pressure when error is zero (Psig)

KC = controller gain (Psig/Psig)

e = error signal = $C_{set} - C$

C_{set} = set point signal

C = process signal from transmitter

It is generally assumed when considering control system dynamics that the control system is at steady state for $t < 0$ and that $e = 0$ for $t < 0$. Hence with proportional control, the greater the magnitude of the error, the larger the corrective action applied. Almost all commercial

controllers are calibrated in terms of proportional band instead of gain.

100

P B=

KC (6.2)

The higher or wider the PB, the lower is the gain and vice versa. For a proportional controller, the transfer function B(s) is

M

$B(s) = \frac{M}{E(s)} = KC$

E (6.3)

Where M and E represents the Laplace transform of the controller output and error respectively

6.1.2 Integral (Reset) Action

The integral and derivative models are normally used in combination with proportional model. Integral action gives an output which is proportional to the time integral of the error. If there is no error, the controller for a pure integral controller is

$M(s) = \frac{1}{s} E(s)$

$B(s) = \frac{M(s)}{E(s)} = \frac{1}{s}$

$\tau_I s$ (6.4)

While its equation is represented by

45

t

1

$M(t) = M + \int_0^t e(t) dt$

τ_I (6.5)

Where τ_I is the integral or reset (time)

6.1.3 Derivative Action

The purpose of the derivative action is to anticipate where process is heading by looking at the time rate of change of the error or its derivative.

de

$M(s) = M + \tau_D s \frac{dE(s)}{ds}$

dt (6.6)

Where τ_D is the derivative time (mins)

In theory, derivative action should always improve dynamic response and it does in many loops. In others, however, the noise problem makes the use of the derivative action controller undesirable. The transfer function for an ideal derivative controller is

M

$B(s) = \tau_D s$

E (6.7)

However, this ideal derivative action cannot be realised, since units (lead – lag) employed in derivative action are only approximate. In the Laplace domain, their transfer function is

$$M \tau D s + 1$$

$$(s) =$$

$$E \alpha \tau D s + 1 \quad (6.8)$$

Where α for commercial controllers is in the range 1/6 to 1/20

6.2 Reasons for Particular Use of Control mode – Offset

No control

Controller

Variable

P

P+I offset

P+D

PID

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Time

The above diagram shows the typical response of a controlled variable to a step disturbance in load. The effect of different control action can be summarised as follows

6.2.1 Proportional Control

The response has a high maximum deviation and there is a significant time of oscillation. The period of this oscillation is moderate. For a sustained change in load, the controlled variable is not returned to its original value (set point) but attains a new equilibrium value termed the control point. The difference between the set point and the control point is called the offset or droop.

6.2.2 P + I Control

From the PI equilibrium, it can be seen that the controller output will continue to increase as long as $e > 0$ (with PI control). The contribution from the integral action does not return to zero with the error, but remain at the value it has reached at that time. With PI control, an error (offset) had to be maintained in order to keep the level constant after a step change in inlet flow rate. This contribution provides an additional output pressure necessary to open valve to keep the level at the set point. No offset is now necessary. Thus, with integral action, offset is eliminated. The disadvantages of PI control are that it gives rise to a higher maximum deviation, a longer response time and a longer period of

oscillation than with proportional action alone. This type of action is therefore used when the above can be tolerated and offset is undesirable. It is frequently used in combination especially where the response of the other components are rapid.

6.2.4 P + D Control

Control action due to the derivative mode occurs only when the error is changing. The presence of the derivative action contributes an additional output to the final element as soon as any change in error occurs. When the error ceases to change, derivative action no longer occurs. The effects of this is similar to having a proportional controller with a high gain when the measured variable is changing most rapidly and low gain when the latter is varying slowly. Due to this action, the controlled variable exhibits the least oscillation and lowest minimum deviation. Derivative action is frequently termed anticipatory or rate control action and is employed where excessive oscillation have to be eliminated.

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6.2.5 PID Control

This is essentially a compromise between the advantages and disadvantages of proportional-integral (PI) control and proportional-derivative (PD) control. Offset is eliminated by the presence of integral action and the derivative mode reduces the maximum deviation and time of oscillation, although the later are still greater than the PI control alone. Such controllers are usually installed of their versatility and not because analysis of the system has indicated the need for the presence of all three mode of control.

7.0 Transient Response of Control Systems

The term transient response could mean the response of a control system to any type of input, but it usually refers to a step change in the set point or a load variable. A step change is used partly for convenience as its solutions are easier to obtain than for any other type of disturbance. The step change is also the most severe type of disturbance, and the response to a step change shows the maximum error that could occur for a given eventual change in load. If several control systems or controller settings are being compared, the system with the best response to a step change to load variable will generally have the best response to random fluctuations of the variable. As far as stability is concerned, it does not matter which input is changed or what type of change is made as long as the system is linear – a closed loop system is unstable to all inputs.

7.1 General Equations for Transient Response

The general equations for the response of a closed system are obtained

by the algebraic manipulations of the signals and transfer functions of the block diagram. The figure below shows a single-loop system with one load variable and no measurement lag.

$\theta_L GL$

$\theta_C + e$

$G \theta$

Set point + Output

For this system

$$\theta = \theta_C G + \theta_L GL, e = \theta_C - \theta$$

(7.1)

or

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$$\theta_C G \theta_L GL$$

$$\theta = \theta_C G - \theta_G + \theta_L GL = +$$

$$1 + G \quad 1 + G$$

(7.2)

The variable θ_C , θ_L and θ are changes from the normal values of the set point, load variables and process output, and they are all initially zero.

The effect of changes in one variable (θ_C , θ_L) can be obtained by setting the other to zero and solving for the specific case. Note that we are neglecting both measurement lag and the controller lag in the following cases and we have the transfer functions for an ideal controller

7.3 Proportional Control of a Single Capacity Process

Consider the case where a first order system and a proportional controller are the only elements in the control system. The time constant for load changes is the same as for changes in the manipulated variable.

θ_L

K_L

$\frac{1}{s} \frac{1}{\tau s + 1}$

$\theta_C + K_p$

$G \theta$

$\frac{1}{s} \frac{1}{\tau s + 1} +$

The valve gain is combined with the process gain K_P and the valve dynamics are neglected. For a change in set point

(())

$\theta K \frac{1}{s}$

$$\theta_C \frac{\tau s + 1}{s + 1} \tau s + 1 + K$$

$\tau s + 1$

$s + 1$

(7.3)

Where $K = K_L K_P$

If $\theta_c = A$, then $\int (\theta_c) = A/s$

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$\theta =$

A

$(K$

$s\tau s + 1 + K)$

(7.4)

Putting equation (6.4) in a familiar form, the numerator and denominator are divided by $K+1$

$(\quad)$

$AK + 1$

$\theta =$

$1 + K\tau$

$s\tau s + 1$

$1 + K$

(7.5)

or

$\theta =$

$AK + 1$

$1 + K$

1

$s(\tau s + 1)$

(7.6)

where

τ

$\tau' =$

$K + 1$

In equation (7.5) θ , is a transformed variable and equation (8.6) is the actual output. The solution of equation (7.6) is

$(\quad)$

$-t$

AK

$\theta = 1 - e^{-\tau' t}$

$1 + K$

(7.7)

Equation (7.7) has the same form as the solution for a step input to an open loop first order system, but the effective time constant is lower than

τ by a factor of $1+K$. Note that the speed of response depends on the overall gain K , which is the product of the process gain. The response to changes in load is obtained from the second term.

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$\theta_{GL\ KL}$

$=$

$\theta_L \frac{1}{1 + G\tau s + 1 + K}$

(7.8)

Where $K = K_C K_P$

If $\theta_L = A$, then $\int (\theta_L) = A/s$

$A \frac{K_L}{s}$

$\theta =$

$\frac{A}{s} \frac{K_L}{\tau s + 1 + K}$

(7.9)

Again the factor $1+K$ is used to reduce the equation to a familiar form

$(\)$

$A \frac{K_L}{1}$

$\theta =$

$\frac{1 + K}{\tau s + 1}$

$s s + 1$

$1 + K$

(7.10)

$A \frac{K_L}{1}$

$(1 - e^{-t})$

$-t$

$\theta = \tau'$

$1 + K$

(7.11)

The effective time constant $\tau' = \tau(1+K)$ is the same for load changes as for changes in set point.

7.4 Proportional Control of Capacity Process

K_L

θ_L

$\frac{1}{2s + 1}$

$+ K_1 K_2$

$\theta_C K_C \theta$

$\frac{1}{s} \frac{1}{1 + 251}$

$s \frac{1}{1 +}$

For the system above, the response to a unit change in load will be

obtained. The process could be a jacket kettle where the temperature is controlled by adjusting the flow of coolant to the jacket. The load variable might be the flow rate or temperature of the process stream fed to the kettle. The time constant of the kettle τ_2 is the same for load changes and changes in jacket temperature, but the gains K_L and K_2 are different

$$\begin{aligned} \theta_G &= L = \frac{K_L}{(\tau_1 s + 1)(\tau_2 s + 1)} \\ \theta_{L+G} &= \frac{K_L \tau_2 s + 1}{(\tau_1 s + 1)(\tau_2 s + 1) + K} \end{aligned} \quad (7.10)$$

Where $K = K_1 K_2$

For $\theta_L = 1$

$$\begin{aligned} & \left(\frac{1}{K_L} \right) \frac{1}{(\tau_1 s + 1)} \\ \theta &= \frac{s \tau_1 \tau_2 s^2 + (\tau_1 + \tau_2) s + (1 + K)}{(\tau_1 s + 1)(\tau_2 s + 1)} \end{aligned} \quad (7.11)$$

The term $K+1$ is used as a factor to make the denominator of equation (7.11) fit the standard form for second order equations

$$\begin{aligned} & \left(\frac{1}{K_L} \right) \frac{1}{(\tau_1 s + 1)} \\ \theta &= \frac{K + 1}{K + 1} \frac{\tau_1 \tau_2 s^2 + (\tau_1 + \tau_2) s + 1}{s^2 + s + 1} \end{aligned} \quad (7.12)$$

$$\theta = \frac{K_L \tau_1 s + 1}{(\tau_1 s + 1)}$$

$$\begin{aligned} & \left(\frac{1}{K + 1} \right) \frac{s^2 + 2 \xi \omega_n s + \omega_n^2}{s^2 + s + 1} \end{aligned}$$

$$\omega_n^2 = \frac{K + 1}{\tau_1 \tau_2} \quad (7.13)$$

Where

$$\begin{aligned} \omega_n &= \sqrt{\frac{K + 1}{\tau_1 \tau_2}} \end{aligned}$$

$$\tau_1 + \tau_2 + 1 + R$$

$$\xi = \frac{R}{2\sqrt{K+1}}$$

$$\tau_1 = \frac{1}{\omega \sqrt{1-\xi^2}}$$

$$\tau_2 = \frac{R}{\omega^2}$$

$$R = 2\xi\omega$$

$$\tau_1 = \frac{1}{\omega \sqrt{1-\xi^2}}$$

The inverse transform of this equation is obtained from the tables; the form of the solution depends on the value of the damping coefficient.

For $\xi < 1$

$$\left[\frac{1}{\omega \sqrt{1-\xi^2}} \right]$$

$$1 - \left(\frac{2\xi\omega\tau_1 + \omega\tau_1}{\omega^2} \right) - \xi\omega t$$

$$2\omega^2$$

$$KL$$

$$\theta = 1 + e \sin(\omega \sqrt{1-\xi^2} t + \phi)$$

$$K + 1(1-\xi^2)$$

$$2$$

$$(7.14)$$

Where

$$\omega\tau_1 \sqrt{1-\xi^2} = \frac{1}{\omega \sqrt{1-\xi^2}}$$

$$\phi = \tan^{-1} \frac{1-\xi^2}{\xi}$$

$$1 - \xi\omega\tau_1 = \xi$$

For $\xi = 1$, the quadratic term becomes $(s/\omega - 1)^2$ or $(\tau_1 s + 1)^2$ and the solution is

$$KL$$

$$\left[(\tau_1 - \tau_a) \right]$$

$$)]$$

$$-t$$

$$\theta = 1 + t - 1 e^{-\tau_1 a}$$

$$K + 1 \tau_1^2 a$$

$$(7.15)$$

Where

$$1 \sqrt{\tau_1^2 + \tau_2^2}$$

$$\tau_a = \frac{1}{\omega^2} = \frac{1}{\omega^2}$$

$$\omega K + 1 \tau_1 + \tau_2$$

$$53$$

For $\xi < 1$, the quadratic term is factored to $(\tau_1 s + 1)(\tau_2 s + 1)$ and the solution is

$$\theta =$$

KL

$K + 1 [()$

$\tau - \tau_1 - t \tau$

$1 + a$

$\tau b - \tau a$

e

$\tau - \tau_1 - t \tau$

$- b$

$1a$

$\tau b - \tau a$

e

) $1b$

(7.16)

The time constant τ_1 and τ_2 are equal. The response is always oscillatory

since $\xi < 1$ for

$\tau_1 = \tau_2$, then

1

$\xi =$

$\sqrt{K + 1}$

For other ratios of τ_2 to τ_1 the response may be under-damped depending on the value of K. Increasing the controller gain decreases the damping coefficient and also decreases the steady state offset. The figure below shows typical results for under-damped and critically damped cases for an example where $\tau_2/\tau_1 = 0.25$

1.0

0.8 $K = 0.56$

$\xi = 1$

0.6

KL

0.4 $K = 4$

$\xi = 0.56$

0.2 $K = 20$

$\xi = 0.27$

0

0.5 1.0

-0.2 Limit

7.5 Stability

t

A closed loop system with three or more first order elements in series is unstable if the overall gain exceeds a certain value. The transfer function

for the closed loop response to changes in set point or load are shown to be

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$$\theta = \frac{G_C \theta_L}{1 + G_C}$$

= ; =

$$\theta = \frac{G_C \theta_L}{1 + G_C}$$

(7.17)

For any changes in θ_C or θ_L , the equation for θ can be written in general term as follows

$$A_m S^m + A_{m-1} S^{m-1} + \dots + A_1 S + A_0$$

$\theta =$

$$\frac{B_n S^n + B_{n-1} S^{n-1} + \dots + B_1 S + B_0}{A_m S^m + A_{m-1} S^{m-1} + \dots + A_1 S + A_0}$$

(7.18)

Where n and m are integers and A and B are constants.

The denominator can be factored to show the roots of the polynomial, which may be real, imaginary, complex numbers or zero

$$A_m S^m + A_{m-1} S^{m-1} + \dots + A_1 S + A_0$$

$\theta =$

$$(S - r_1)(S - r_2) \dots (S - r_n)$$

(7.19)

The inverse transform yields terms containing $e^{r_1 t} \dots e^{r_n t}$. If r_1 is a real negative number, the term containing $e^{r_1 t}$ decays to zero. If r_1 and r_2 are $e^{(x \pm iy)t}$, which decays to zero if the real part x is negative. If any of the roots is real or has a real positive part, the term containing that root eventually becomes very large and the system is unstable.

A control system is stable if the denominator of the closed loop transfer function has no positive roots or no roots with positive real parts. The denominator of the transfer function set equals to zero is the characteristic equation. The maximum gain can be found by factoring the characteristic equation for various gains or by determining the conditions that just make one of the roots positive.

For proportional control of a process with three time constants, the denominator of the closed loop transfer function is

$$K + (\tau_1 s + 1)(\tau_2 s + 1)(\tau_3 s + 1)$$

(7.20)

which can be factored into a first order term and a quadratic term. When the gain exceeds the critical value, the damping coefficient in the quadratic term becomes negative or the two complex roots acquire

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positive real parts. The maximum gain is found by setting $\xi = 0$ in the following equation

$$\left(\frac{s^2 + 2\xi\omega_n s + \omega_n^2}{K + (\tau_1 s + 1)(\tau_2 s + 1)(\tau_3 s + 1)} \right) = \frac{(\tau_1 s + 1)^2 + \omega_n^2}{s + 1(K + 1)} \quad (7.21)$$

By equating the coefficients and using the ratios of time constants, the maximum gain and critical frequency are shown to be

$$1 + K_{\max} = \frac{1 + R_2 + R_3}{1 + R_2 R_3} \quad (7.22)$$

$$\omega_C = \frac{1}{\tau_1 R_2 + R_3 + R_2 R_3} \quad (7.23)$$

Where

$$R_2 = \frac{\tau_2}{\tau_1} \\ R_3 = \frac{\tau_3}{\tau_1}$$

The stability of the system depends on the overall gain and the ratios of the time constants. The maximum gain becomes very large if any one of the three time constants is either much larger or much smaller than the others.

7.6 Effects of Time Delay

In many process control systems, there is a time delay between elements of the control loop. A time delay exists if the temperature of a stream leaving a heat exchanger is measured a few metres downstream from the end of the heat exchanger.

Analytical instruments for reactors or distillation columns are usually placed in a separate simple loop and there is a time delay before changes in system concentrations are transmitted to the instrument cell. With plug flow reactors, the time delay is equal to the residence time in the reactor. The terms dead time and transportation lag also refer to pure time delay.

A time delay in the control loop is represented by a block with transfer

function e^{-DS} . Frequency response analysis shows that time has a

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destabilising effect on the control systems, more so than adding a time constant of the same magnitude. The transient response of the system is not easily obtained because the number of roots is infinite as shown by the expression of the exponential term

$$e^{-DS} = 1 - DS + \frac{(DS)^2}{2!} - \frac{(DS)^3}{3!} + \dots$$

$$e^{-DS} = 1 - DS + \frac{(DS)^2}{2!} - \frac{(DS)^3}{3!} + \dots$$

$$(7.23)$$

The appropriate transient response could be obtained by using a few terms from the above series, but other approximation for the time delay are easier to handle. The following PADE approximation is simple to use and often enough.

$$e^{-DS} \approx \frac{1 + DS/2}{1 + DS}$$

$$(7.24)$$

$$(7.24)$$

$$(7.24)$$

8.0 Frequency Response Analysis

Frequency response analysis means the response of a single element to sinusoidal input, covering a wide range of frequencies

The advantage of using the frequency response to analyse or design control systems is that the system response is obtained easily from the response of the individual elements, no matter how many elements are included. By contrast, calculations of the transient response are quite tedious with only three components in the system, and are too difficult to be worthwhile for four or more components. The basic results obtained from a frequency response analysis are values of the maximum gain for stable operation and the critical frequency of the system. The advantages of frequency response analysis can be summarised as follows:

1. Frequency response is obtained without actually integrating the differential equation. The response of a complete system can be derived from the responses of individual elements of the system by simple graphical constructions.

2. Some systems found in process control present a difficult problem because it is almost impossible to accurately estimate the system parameters. In such cases, the frequency responses may be

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determined experimentally (the use of sine wave generators to

inject sinusoidal signal into either a closed loop or open loop)

8.1 Review of Complex Numbers

The frequency response analysis employs the use of complex numbers. A complex number is usually written in the form

$$Z = x + iy$$

(8.1)

Where Z and i are the co-ordinates representing the numbers and with $i = \sqrt{-1}$

The complex number represents a vector which has a component $+y$ on the imaginary axis

y

θ $x + \text{Re}$

Imagin

The magnitude of the vector is $\sqrt{x^2 + y^2}$ and angle θ is $\tan^{-1}(y/x)$. The number may be written in polar form or in exponential form as

$$Z = Z \angle \theta = \sqrt{x^2 + y^2} \angle \theta$$

y

$= x$

(8.2)

$$Z = Z e^{i\phi}$$

(8.3)

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To add complex numbers, the real and imaginary parts are combined separately after putting the numbers in the form of equation (8.1)

$$Z_1 + Z_2 = (x_1 + iy_1) + (x_2 + iy_2) = (x_1 + x_2) + i(y_1 + y_2)$$

(8.4)

To get the product of two complex numbers, the magnitudes are multiplied and the phase angles added as shown by multiplying the numbers in exponential form

$$Z_1 Z_2 = Z_1 e^{i\phi_1} Z_2 e^{i\phi_2}$$

$$Z_1 Z_2 = Z_1 Z_2 e^{i(\phi_1 + \phi_2)}$$

(8.5)

The conjugate of a complex number is obtained by changing the sign of the imaginary part. Conjugate numbers are used in clearing complex fractions

$$\frac{1}{x + iy} \cdot \frac{x - iy}{x - iy}$$

$=$

$$\frac{x - iy}{x^2 + y^2}$$

(8.6)

8.2 Application of Complex Numbers in determining the Frequency

Response of a System

The frequency response of an element or an entire system can be obtained directly from the transfer function without performing the inverse transformation or otherwise integrating the corresponding differential equation. If we transform, variable, s is replaced by $j\omega$. The resulting complex number shows the amplitude and phase angle corresponding to a sinusoidal input with frequency in radians/unit time.

Let us consider a first order system

1

$$G(s) =$$

$$\tau s + 1$$

(8.7)

On transformation we have

1 1

$$G(j\omega) =$$

$$\tau(j\omega) + 1 \quad j\omega\tau + 1$$

(8.8)

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To eliminate $(j\omega)$ from the denominator, we take the conjugate of the function

$$1 \quad 1 - j\omega\tau \quad 1 - j\omega\tau$$

$$* =$$

$$j\omega\tau + 1 \quad 1 - j\omega\tau \quad 1 + \omega^2 \tau^2 \quad (8.9)$$

Which can be written as

$$1 \quad j\omega\tau$$

$$1 + \omega^2 \tau^2 \quad 1 + \omega^2 \tau^2$$

$$2 \quad 2$$

(8.10)

The transfer function can be written as

$$[]$$

$$1$$

$$y(s) \quad y(j\omega) \quad 1 \quad \omega^2 \tau^2 \quad 2 \quad 2$$

$$= = +$$

$$x(s) \quad x(j\omega) \quad 1 + \omega^2 \tau^2 \quad 1 + \omega^2 \tau^2$$

(8.11)

$$[] \sqrt{\quad}$$

1

$$1 + \omega^2 \tau^2$$

$$\frac{1}{1 + \omega^2 \tau^2}$$

$$(1 + \omega^2 \tau^2)^{-1} \quad (8.12)$$

The phase angle is given by

$$\Phi = \tan^{-1}(-\omega\tau) \quad (8.13)$$

Φ = phase lag (change in shift)

ω = angular frequency

For a first order response, we have

KP

$$\frac{K}{s + 1}$$

KP

$$G(s) =$$

$$\frac{K}{\tau s + 1} \quad (8.14)$$

When transformed

KP

$$G(j\omega) =$$

$$\frac{K}{j\omega\tau + 1} \quad (8.15)$$

Multiplying by the conjugate

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$$\frac{K}{1 + j\omega\tau} \cdot \frac{K}{1 - j\omega\tau}$$

$$\frac{K^2}{(1 + j\omega\tau)(1 - j\omega\tau)} = \frac{K^2}{1 + \omega^2 \tau^2} \quad (8.16)$$

i.e.

$$\frac{K}{1 + \omega^2 \tau^2}$$

$$G(j\omega) = \frac{K}{1 + \omega^2 \tau^2}$$

$$\frac{K}{1 + \omega^2 \tau^2} \quad (8.17)$$

The aptitude ratio, A.R is the magnitude of the transfer function

$$\sqrt{\frac{K^2}{1 + \omega^2 \tau^2}}$$

$$\frac{K}{1 + \omega^2 \tau^2}$$

$$\frac{K}{1 + \omega^2 \tau^2}$$

$$A.R = |G(j\omega)| = \frac{K}{1 + \omega^2 \tau^2}$$

$$\frac{K}{1 + \omega^2 \tau^2}$$

$$= \frac{K}{1 + \omega^2 \tau^2}$$

$$A.R = \frac{K}{1 + \omega^2 \tau^2} \quad (8.18)$$

Or

$$\sqrt{\frac{K^2}{1 + \omega^2 \tau^2}}$$

$$A.R = \frac{K}{1 + \omega^2 \tau^2}$$

$$\frac{K}{1 + \omega^2 \tau^2} \quad (8.19)$$

$$-K P \omega \tau / (1 + \omega^2 \tau^2)$$

$$\angle G(j\omega) = \tan^{-1} \frac{-\omega \tau}{1} = -\tan^{-1} \omega \tau$$

$$K P / (1 + \omega^2 \tau^2) \quad (8.20)$$

KP

$$\therefore G(j\omega) = 1$$

$$\angle \tan^{-1}(-\omega \tau)$$

$$(1 + \omega^2 \tau^2)^{-1/2} \quad (8.21)$$

Second Order Systems

The frequency response for a second order element or system is found by substituting $j\omega$ for s in the general equation for underdamped systems

$$\theta^2 + 2\zeta\omega_n\theta + \omega_n^2 = 0$$

$$\theta = \frac{-2\zeta\omega_n \pm \sqrt{(2\zeta\omega_n)^2 - 4\omega_n^2}}{2}$$

$$\theta = -\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2}$$

$$s = -\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2}$$

$$+1 \pm j\sqrt{1 - \zeta^2}$$

$$\omega_n$$

$$2$$

$$\omega_n^2$$

$$(8.22)$$

$$\omega_n^2 \pm j\omega_n\sqrt{1 - \zeta^2}$$

$$1 \pm j\sqrt{1 - \zeta^2}$$

Multiplying numerator and denominator by

$\omega_n^2 \pm j\omega_n\sqrt{1 - \zeta^2}$ gives

$$\theta^2$$

$$1 - \zeta^2$$

$$\omega_n^2$$

$$\omega_n$$

$$-j$$

$$2\zeta$$

$$\omega$$

$$\omega_n$$

$$((\omega_n^2 \pm j\omega_n\sqrt{1 - \zeta^2}))((\omega_n^2 \pm j\omega_n\sqrt{1 - \zeta^2}))$$

$$\theta^2 + 2\zeta\omega_n\theta + \omega_n^2 = 0$$

$$2$$

$$\omega_n^2$$

$$2$$

$$\omega_n^2$$

$$1 - \zeta^2 \pm j2\zeta\omega_n(1 - \zeta^2)$$

$$\omega_n^2 \pm j\omega_n\sqrt{1 - \zeta^2}$$

$$(8.23)$$

Or

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[()]

$\omega^2 \omega$

$1 - 2\xi$

$\theta^2 \omega_n \omega_n$

$= \angle \tan -$

$\sqrt{(\quad)(\quad)}$

$\theta^2 \omega_n^2 \omega^2$

$\omega^2 \omega^2 -$

$1 - 4\xi \omega_n$

$\omega_n \omega_n$

(7.24)

8. 3 Bode Diagrams

The Bode diagram or plot is a convenient method of presenting response data at various frequencies. Two plots are usually required i.e. a log-log plot for amplitude ratios against the frequency and a semi-log plot of the phase angle against the frequency. The two plots show how magnitude ratio and phase angle vary with frequency. In both plots $\omega\tau$ is plotted as the abscissa. The plots are:

- i. $\log_{10} (AR)$ vs $\log_{10} \omega$
- ii. Φ vs $\log_{10} \omega$

The data for a Bode plot for a first order system is generated as follows:

The transfer function of a first order system is given by:

1

$G(s) =$

$\tau s + 1$

(8.25)

Which in the frequency domain becomes

1

$G(j\omega) =$

$j\omega\tau + 1$

(8.26)

1

The A.R = $\sqrt{1 + \omega^2 \tau^2}$

2 2

(8.27)

$\phi = \tan^{-1}(-\omega\tau)$

(8.28)

1

Thus $\log (A \cdot R) = -\log (1 + \omega^2 \tau^2)$

2

(8.29)

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Limit $\log (A \cdot R) = -\log (\omega \tau)^2$

 $\omega \tau \rightarrow \infty$

(8.30)

Since $(\omega \tau)^2 \gg 1$, $\omega \tau \rightarrow \infty$

The magnitude ratio (A.R) is usually converted to a log modulus defined by the equation

$$L \equiv \log \text{ modulus} \equiv 20 \log |M|$$

(8.31)

The units of log modulus are decibel (db), a term originally used in communication engineering to indicate the ratio of two values of power. The log modulus plotted in Bode plots is often the normalized magnitude ratio. This is obtained by dividing the magnitude ratio by the absolute value of the steady state gain of the transfer function. The normalization is necessary in order to make most of the plots start at zero

 $|M|/|M|$
 $L \equiv 20 \log =$
 $| \text{Steady state gain} | / |K_P|$

(8.32)

but $K_P = \lim_{\omega \rightarrow 0} G(j\omega) = G(0)$

 $\omega \tau \rightarrow 0$

(8.33)

 $|G(j\omega)|$
 $\therefore L \equiv 20 \log =$
 $|G(0)|$

(8.34)

9.3.1 Bode Plots for Some Common Transfer Functions

Gain: If $G(s)$ is just a constant K_P , then

$M = K_P$ and $\phi = 0$. Neither, M or ϕ varies with frequency. The log modulus is

 $\therefore L = 20 \log |K_P|$

(8.35)

Both the phase angle and log modulus curves are horizontal lines. Increasing K_P moves the L curve up while a decrease in K_P moves it

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down. If KP is greater than unity, L is positive and if K P is less than unity, L is negative. The Bode plots are shown below

$$20 \log (KP) \quad KP > 1$$

$$0.1 \quad 1.0 \quad 10 \quad 0.1 \quad 1.0 \quad 10$$

$$\omega$$

1st Order Lag

$$KP$$

$$M =$$

$$\sqrt{1 + \omega^2 \tau^2}$$

$$(8.36)$$

Normalizing by eliminating KP, we have

$$1$$

$$L = 20 \log$$

$$\sqrt{1 + \omega^2 \tau^2}$$

$$(8.37)$$

$$\theta = \tan^{-1}(-\omega\tau)$$

$$(8.38)$$

The Bode plots are shown below

$$LL$$

$$LFA$$

$$0 \quad 0$$

$$HFA$$

$$-6$$

$$\phi\phi - 45$$

$$-12$$

$$-900. \text{ Slope}$$

$$-18 - 20 \text{ db/decade}$$

$$1 \quad 1 \quad 10$$

$$10$$

$$\square 64 \quad \square \quad \square$$

$$\log \omega \quad \log \omega$$

One of the most convenient features of Bode plots is that the L curve can easily be sketched by considering the low and high frequency asymptotes. As ω goes to zero, the normalized L curve goes to 0. As ω becomes very large, equation 8.37 becomes

$$1$$

$$\lim L = 20 \log = -20 \log \omega\tau$$

$$\sqrt{1 + \omega^2 \tau^2}$$

$$\omega \rightarrow \infty$$

(8.39)

$$= -20 \log \omega - 20 \log \tau \quad (8.40)$$

This is the equation of a straight line of L versus ω . It has a slope of -20 . As $\log \omega$ increases by 1 or a factor of 10, L decreases by 20 db. A factor of 10 is called a decade while a factor of 2 is an octave. So the slope of the high frequency asymptote is -20 db/decade or -6 db/octave

The high frequency asymptote intersect the $L = 0$ line at $\omega = 1/\tau$. This is called the break frequency. The log modulus curve is flat (horizontal) up to the break frequency, then begins to drop off. Thus the L curve can easily be sketched by drawing a line with a slope of -20 db/decade from the break frequency which is the reciprocal of τ . The phase angle $= 45^\circ$ at the break frequency

1st Order Lead

The normalized transfer function is given by

$$1$$

$$L = 20 \log$$

$$\sqrt{1 + \omega^2 \tau^2}$$

(8.41)

$$\theta = \tan^{-1}(\omega \tau)$$

(8.42)

The high frequency asymptote has a slope of $+20$ db/decade. The break frequency is $1/\tau$. The phase angle goes from zero to $+90^\circ$ and is equal to $+45^\circ$ at the break frequency. The plots are shown below

Slope

-20 db/decade

L

90.

12.

$\phi = 45^\circ$ HFA

6.

0.

0.

(a) Dead Time

The transfer function for a dead time is

$$G(j\omega) = e^{-j\omega D}$$

(8.43)

$$L = 20 \log 1 = 0 \quad (8.44)$$

$$\phi = -\omega D \quad (8.45)$$

The dead time transfer function has a flat L curve for all frequencies, but the phase angle drops off to minus infinity. The plots are shown below

0

L (db)

- 90

ϕ

0

- 180

$\square \square$

$\square \square \square \text{Log } \omega$

2D D

Log ω

(b) Systems in series

The usefulness of logarithmic plot becomes apparent when it is desired to determine the frequency response of systems in series. The resultant amplitude ratio and phase shift may be obtained as follows:

$$A.R = K \cdot (A.R)_1 \cdot (A.R)_2 \cdot (A.R)_3 \cdot \dots$$

(8.46)

$$\text{Log } (A.R) = \text{log } K + \text{log } (A.R)_1 + \text{log } (A.R)_2 + \text{log } (A.R)_3 + \dots$$

(8.47)

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$$\Phi = \Phi_1 + \Phi_2 + \Phi_3 + \dots$$

(8.48)

The overall Bode plot of a number of systems in series may therefore be obtained as follows"

a) Add all A.Rs to determine the resultant A.R in the log-log plot, treating values above unity A.R as positive and those below unity A.R as negative. It is sufficiently accurate to add the asymptotes

b) Add the individual phase shifts on the linear plot to obtain the resultant phase shift

c) If the resultant transfer function is multiplied by some constant, e.g. the steady state gain, the entire A. R curve is moved vertically by a constant amount. The phase shift is unaffected

(c) System Stability from the Bode Diagrams

The Bode stability criterion states that a control system is unstable if its open loop frequency response exhibits an amplitude ratio greater than unity at the frequency for which the phase shift is -180° . This frequency is termed the cross-over frequency. The degree of stability of the system can be obtained from its Bode diagram. If the open-loop A.R is unity when the phase shift is -180° , the closed loop control system will oscillate with

constant amplitude. The greater the difference between the open-loop A.R (< 1) at ω_c and A.R = 1, the more stable the closed-loop system will be

(d) Effects of Time Delay

The frequency response of a time delay is an amplitude ratio of 1.0 and a phase lag equal to the product of the delay and the frequency. For a time delay of D seconds

$$A.R = 1.0$$

$$\Phi = -\omega D$$

Since the phase lag increases without limit, any process with a time delay has a finite critical (cross-over) frequency and a finite maximum gain.

The controllability can be improved by decreasing the time delay in the system, since the total phase lag is decreased and the overall amplitude ratio is unchanged. This increases the critical frequency which usually results in an increase in the maximum gain, since A.R for most systems decreases with increasing frequency. If there are more than one delay in the control loop, they can be lumped together to calculate the overall response of the system.

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